

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	SWTID-2026-8809
Project Title	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to simplify learning using generative AI, improving understanding and saving study time. It tackles the difficulties students face in self-study, promising clearer explanations, better practice and structured guidance. Key features include AI-based Q&A, concept explanation, quizzes, summaries and learning paths.

Project Overview	
Objective	The primary objective is to simplify learning for students of all levels by providing an AI-powered assistant that answers questions, explains concepts, generates quizzes, summarizes passages and recommends learning paths in one place.
Scope	The project covers five modules (Explanation, Q&A, Quiz, Summary and Learning Path) built with FastAPI and a lightweight HTML + CSS frontend, using Gemini 1.5 Pro through the API and LaMini-Flan-T5 locally.
Problem Statement	
Description	Students often struggle with difficult language in textbooks, scattered online answers, lack of practice quizzes, long notes and no clear learning plan, which makes self-study slow and confusing.
Impact	Solving these issues will give learners faster and clearer understanding, better exam readiness, saved revision time and a structured way to learn, improving confidence and learning outcomes.
Proposed Solution	
Approach	Combining cloud-based generative AI (Gemini 1.5 Pro) with a lightweight local model (LaMini-Flan-T5) behind a modular FastAPI backend, so each learning task has its own endpoint and logic, and results appear instantly on a simple web page.

Key Features	- AI-powered question answering (Q&A) using Gemini 1.5 Pro. - Simple concept explanations using the local LaMini-Flan-T5-783M model. - Quiz generation: 3 MCQs with 4 options each, with correct answer feedback. - Summarization of long passages for quick revision. - Personalized learning paths from beginner to advanced level, with resources. - Simple web interface (HTML + CSS) with real-time results from the FastAPI backend.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	CPU only (e.g., Mac M1); no GPU required
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	5 GB free space
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
AI Models	Cloud and local models	Gemini 1.5 Pro (API), LaMini-Flan-T5-783M (local)
Libraries	Additional libraries	google-generativeai, transformers, torch
Development Environment	IDE	Python 3.10+, VS Code / PyCharm
Data		
Data	Source, size, format	User text input (runtime), no fixed dataset; JSON responses

Team Work Allocation

S.No	Team Member	Work Assigned
1	Divya K	Project Overview (Objective and Scope)
2	Veronica	Problem Statement (Description and Impact)
3	Sangeetha T	Proposed Solution (Approach and Key Features)
4	Pooja N	Resource Requirements: Hardware and Software
5	Monika N	Resource Requirements: Data, and final review of the report